Senior Engineer I, Software Engineering- "Factory Automation"

ASM · Greater Phoenix Area

Job ID (this archive)	MFG-089
Employer	ASM
Location	Greater Phoenix Area
Career stage	Experienced
Location group	Outside Texas
Major (archive tag)	Software Engineering
Sector area	Electronics & semiconductor manufacturing
Industry (LinkedIn)	Semiconductor Manufacturing
Pay listed	Not listed
Stated rules	No GPA or sponsorship rule stated
Posted / closes	2026-10-19 · 2027-04-26
Status when checked	Accepting applications on LinkedIn (Nov 1, 2026)
Source page	https://www.linkedin.com/jobs/view/4295432099/
Captured	2026-11-01

Full posting text as it appeared on LinkedIn on Nov 1, 2026, saved for ECS 1100. Postings change and close often; open the source page above for the current version before applying.

At a glance

What you would do

Software engineer developing and debugging C# control software for semiconductor manufacturing equipment and its communication with factory automation systems.

Who can apply

- **Majors named:** Computer Science
- **Degree and experience:** BS in Computer Science or related (MS preferred); 5+ years software development; semiconductor and factory automation experience

Skills mentioned

Teamwork & collaboration · Communication · Problem solving & root cause · Automation & controls · Programming

Full posting text

Step into a career with ASM, where cutting edge technology meets collaborative culture.

For over 55 years ASM has been ahead of what’s next, at the forefront of innovation and what’s technologically possible. With more than 4,500 ASMers representing 70 nationalities, our people and our advanced semiconductor devices are playing a crucial role in trends such as 5G, cloud computing, AI, and autonomous driving. But we’re more than just a tech company. We value diversity, inclusion and sustainability as we strive to make a positive impact on the world. Our development programs help support your growth, shaping your future and pushing the boundaries of innovation to unleash potential.

Senior Engineer I, Software Engineering - Factory Automation

As part of ASM's Software Engineering team, you'll develop advanced control software that drives the most sophisticated semiconductor equipment in the world. Your work will directly impact robotics, gas flow systems, and temperature control hardware—delivering real-time results in chip manufacturing. Based in Phoenix, you'll collaborate globally and help shape the future of semiconductor automation and factory integration.

Responsibilities

Design, develop, test, deploy, and maintain software for semiconductor manufacturing equipmentBuild software for Windows and real-time operating systemsTroubleshoot and resolve issues in real-time control systemsTriage mechanical, electrical, and software bugsLead escalation calls with field engineers and global customersCollaborate with QA and Support teams to manage regression testingAuthor and review system specifications, software designs, and test plansInvestigate internal and external problem reports and propose solutionsEvaluate and recommend software engineering toolsParticipate in system design and scheduling effortsIntegrate and support factory automation protocols such as SECS/GEMWork closely with automation teams to ensure seamless equipment-factory communication

Requirements

BS Degree in Computer Science or related field. MS preferred.Strong software development skills in C#5+ years' experience as a generalist or mid-stack software developerExperience in the semiconductor industryHands-on experience with factory automation systemsFamiliarity with SECS/GEM, GEM300, EDA/Interface A protocolsExperience with AMHS (Automated Material Handling Systems)

What Sets You Apart

10+ years in software development and architectureExperience with:Machine control software (especially for semiconductor equipment)Windows and real-time operating systemsMulti-threaded and real-time applicationsGUI development and device driversNetworking, data communications, and EtherCATDynamic scheduling algorithms with deadlock avoidance

Apply Today To Be Part Of What's Next.

We make the tech that enables the chips in devices which improve lives around the world. We do this with an eye to the future, pushing the boundaries of what's possible through cutting-edge innovation, and driving the next wave of technological breakthroughs that shape how we live, work, and connect.

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